

Unsupervised ECG Anomaly Detection Using an LSTM Autoencoder

A Recording-Level Study of Atrial Fibrillation in the Georgia Dataset

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Abstract

This report presents the implementation and evaluation of a reconstruction-based anomaly detection pipeline for 12-lead electrocardiogram (ECG) recordings from the Georgia subset of the PhysioNet/Computing in Cardiology Challenge 2020 dataset. The objective is to investigate whether an LSTM autoencoder can identify atrial fibrillation (AF) recordings through reconstruction error without using diagnostic labels during model optimization.

The implemented pipeline includes dataset validation, recording-level partitioning to prevent data leakage, signal normalization, sliding-window generation, LSTM autoencoder training, hierarchical reconstruction-error aggregation, and validation-based threshold selection. Two training protocols are compared: Protocol A trains on both normal and AF recordings, whereas Protocol B follows a one-class strategy by training exclusively on normal recordings.

Experimental results demonstrate that Protocol B provides better discrimination between normal and AF recordings, achieving a test ROC-AUC of 0.6851 and a PR-AUC of 0.6138, compared with 0.6413 and 0.5751 for Protocol A. Although validation-based threshold optimization improves the recording-level F1-score, considerable overlap remains between normal and AF reconstruction scores. Overall, the proposed implementation establishes a complete, reproducible, and technically sound baseline for reconstruction-based ECG anomaly detection.

Keywords: ECG, anomaly detection, LSTM autoencoder, Atrial Fibrillation, reconstruction error, unsupervised learning.

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1 Introduction

This project implements and evaluates a reconstruction-based anomaly detection pipeline for identifying atrial fibrillation (AF) from 12-lead electrocardiogram (ECG) recordings using an LSTM autoencoder. The implementation follows the methodology proposed in the reference work provided with the assignment while adapting the complete pipeline to the Georgia subset of the PhysioNet/Computing in Cardiology Challenge 2020 dataset.

Unlike supervised ECG classification, the proposed approach does not use diagnostic labels during model optimization. Instead, the autoencoder learns to reconstruct ECG signals, and reconstruction errors are used as anomaly scores. AF labels are employed only for constructing the experimental splits and evaluating whether abnormal recordings produce higher reconstruction errors than normal recordings.

Beyond implementing the neural network, this work develops a complete and reproducible machine learning pipeline, including dataset validation, recording-level partitioning to prevent information leakage, signal normalization, sliding-window generation, model training, hierarchical reconstruction-error aggregation, threshold selection, and recording-level evaluation.

The implementation investigates two experimental protocols. Protocol A trains the autoencoder using both normal and AF recordings without providing labels to the model, whereas Protocol B follows a one-class anomaly detection strategy by training exclusively on normal recordings. Comparing these protocols allows the effect of the training data composition on reconstruction-based anomaly detection to be evaluated under an identical preprocessing and evaluation framework.

The main objectives of this work are to:

1. implement a complete reconstruction-based ECG anomaly detection pipeline for the Georgia dataset;
2. compare mixed-class and normal-only training strategies;
3. investigate hierarchical aggregation methods for converting window-level reconstruction errors into recording-level anomaly scores; and
4. evaluate the robustness of the resulting detector using validation-based threshold selection and independent test data.

2 Problem Formulation

Let an ECG recording be denoted by R_i , where each recording consists of one or more signal segments,

$$R_i = \{S_{i,1}, S_{i,2}, \dots, S_{i,m_i}\}.$$

Each segment is represented as a multivariate time series,

$$S_{i,j} \in \mathbb{R}^{2500 \times 12},$$

where 2500 denotes the number of temporal samples and 12 represents the standard ECG leads.

Each recording is associated with a binary diagnostic label,

$$y_i = \begin{cases} 0, & \text{Normal,} \\ 1, & \text{Atrial Fibrillation (AF).} \end{cases}$$

Rather than predicting y_i directly, the objective is to learn a reconstruction model

$$\hat{X} = f_{\theta}(X),$$

where X denotes an input ECG window and $f_{\theta}(\cdot)$ is an LSTM autoencoder parameterized by θ . Model parameters are optimized by minimizing the reconstruction loss,

$$\theta^* = \arg \min_{\theta} \frac{1}{N} \sum_{n=1}^N \mathcal{L}(X_n, \hat{X}_n).$$

During inference, reconstruction errors are aggregated to obtain a recording-level anomaly score $A(R_i)$. The central hypothesis of this work is that recordings labelled with AF produce higher reconstruction errors than normal recordings, enabling AF detection without using diagnostic labels during model optimization.

3 Dataset

The implementation uses the Georgia subset of the PhysioNet/Computing in Cardiology Challenge 2020 dataset. The original signal repository contains 10,344 recording identifiers and 20,636 ECG segments. After matching signal identifiers with the available metadata and retaining only recordings with valid AF annotations, the final dataset consists of 3,000 ECG recordings and 5,983 signal segments.

Table 1: Dataset summary after metadata matching and validation.

Property	Value
Original recordings	10,344
Original segments	20,636
Retained recordings	3,000
Retained segments	5,983
ECG leads	12
Samples per segment	2,500

Before preprocessing, all retained recordings were validated to ensure data consistency. Each ECG segment was verified to contain 12 leads and 2,500 samples, with no missing or non-finite values. Signal orientation was converted from (12, 2500) to (2500, 12) to match the sequence-first input format required by the LSTM autoencoder.

The anomaly detection task focuses on identifying recordings labelled with atrial fibrillation (AF). Consequently, the negative class represents recordings *without AF* rather than clinically healthy subjects. This distinction is important because non-AF recordings may still contain other cardiac abnormalities or acquisition artefacts that increase reconstruction error.

4 Data Pipeline

Figure 1 summarizes the complete implementation pipeline developed in this project. Starting from the validated ECG recordings and metadata, the workflow performs recording-level partitioning, training-only normalization, sliding-window generation, LSTM autoencoder training, hierarchical reconstruction-error aggregation, and recording-level anomaly detection.

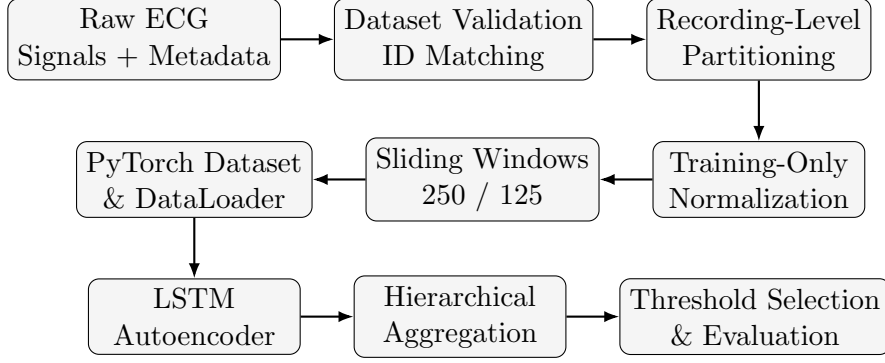


Figure 1: Overall implementation pipeline adopted in this work.

4.1 Recording-Level Partitioning

To prevent information leakage, dataset partitioning is performed at the recording level before preprocessing and window generation. Consequently, all segments and windows originating from the same ECG recording remain within a single data split.

Both protocols use identical balanced validation and test sets, each containing 76 Normal and 76 AF recordings. Their training strategies differ, as summarized in Table 2.

Table 2: Recording-level composition of the experimental protocols.

Protocol	Training strategy	Training recordings	Validation / Test
A	Normal + AF (labels hidden)	1,068	152 / 152
B	Normal only	2,340	152 / 152

Protocol A learns from both Normal and AF recordings without using diagnostic labels during optimization, whereas Protocol B follows a one-class anomaly detection strategy by training exclusively on Normal recordings.

4.2 Signal Preprocessing

For each protocol, lead-wise normalization statistics are estimated exclusively from the corresponding training set and then applied unchanged to the validation and test sets. This prevents evaluation data from influencing the preprocessing stage.

Each normalized ECG segment is subsequently divided into overlapping windows of 250 samples using a stride of 125, producing 19 windows per segment. This representation increases the number of training samples while preserving local temporal information.

Table 3: Dataset size after window generation.

Protocol	Split	Recordings	Segments	Windows
A	Training	1,068	2,129	40,451
A	Validation	152	304	5,776
A	Test	152	303	5,757
B	Training	2,340	4,665	88,635
B	Validation	152	304	5,776
B	Test	152	303	5,757

5 LSTM Autoencoder

The model receives normalized ECG windows of shape 250×12 , where 250 is the sequence length and 12 is the number of ECG leads. Its objective is to reconstruct each input window without using diagnostic labels during optimization.

The encoder processes the complete sequence using a single-layer LSTM with hidden size 128. The final hidden state is regularized with dropout and projected into a 64-dimensional latent representation. The decoder repeats this latent vector across 250 time steps, processes the resulting sequence with a second LSTM of hidden size 128, and applies a linear output layer to reconstruct the original 12 leads at every time step.

The complete model contains 222,924 trainable parameters. Table 4 summarizes the architecture and training configuration.

Table 4: LSTM autoencoder and training configuration.

Configuration	Value
Input shape	250×12
Encoder hidden size	128
Latent dimension	64
Decoder hidden size	128
Recurrent layers	1 per LSTM block
Dropout	0.20
Trainable parameters	222,924
Loss function	Smooth L1, $\beta = 0.5$
Optimizer	AdamW
Initial learning rate	10^{-3}
Weight decay	10^{-5}
Batch size	64
Maximum epochs	50
Gradient clipping norm	1.0
Early-stopping patience	8
Learning-rate scheduler	ReduceLROnPlateau

Training minimizes the mean Smooth L1 reconstruction loss over all samples, time steps, and ECG leads. For a reconstruction difference $d = x - \hat{x}$, the element-wise loss is

$$\ell(d) = \begin{cases} \frac{d^2}{2\beta}, & |d| < \beta, \\ |d| - \frac{\beta}{2}, & |d| \geq \beta. \end{cases}$$

Smooth L1 loss was selected because it combines stable optimization for small reconstruction errors with reduced sensitivity to isolated high-amplitude ECG samples. AdamW optimization, gradient clipping, learning-rate reduction, and early stopping were used to improve training stability. The best checkpoint for each protocol was selected according to validation reconstruction loss.

6 Anomaly Scoring and Aggregation

The autoencoder produces a reconstructed window $\hat{X}_{i,j,k}$ for every input window $X_{i,j,k}$. The implementation computes Smooth L1 error, mean absolute error (MAE), and mean squared error (MSE) at the window level.

Because the final decision is made for complete ECG recordings, window-level errors must be aggregated hierarchically. Several combinations of window score, window-to-segment aggregation, and segment-to-recording aggregation were evaluated using validation data only.

For both protocols, the selected configuration was:

- window anomaly score: MAE;
- window-to-segment aggregation: median;
- segment-to-recording aggregation: maximum.

For segment $S_{i,j}$, the aggregated score is

$$q_{i,j} = \text{median}_k \left[\text{MAE} \left(X_{i,j,k}, \hat{X}_{i,j,k} \right) \right].$$

The anomaly score assigned to recording R_i is then

$$r_i = \max_j q_{i,j}.$$

The median reduces the influence of isolated high-error windows within a segment, while the maximum assigns the recording score according to its most anomalous segment. This is appropriate when a recording contains multiple segments and abnormal behavior may not be present uniformly across all of them.

Table 5: Validation performance of the selected recording-score configuration.

Protocol	Validation ROC-AUC	Validation PR-AUC
A	0.6569	0.6399
B	0.6827	0.6691

The aggregation strategy was fixed after validation selection and subsequently applied unchanged to the training, validation, and independent test recordings.

7 Threshold Selection

Two validation-derived thresholds are used for each protocol.

The primary unsupervised threshold is defined as the 95th percentile of the non-AF validation recording scores:

$$\tau_{95} = Q_{0.95} (\{r_i : y_i = 0, i \in \mathcal{V}\}).$$

This gives thresholds of 0.339208 for Protocol A and 0.307567 for Protocol B. This operating point is intentionally conservative and is expected to favour specificity.

A second threshold is selected by maximizing the F1-score on the validation set:

$$\tau_{F1} = \arg \max_{\tau} F1(\tau).$$

The resulting thresholds are 0.181348 for Protocol A and 0.173709 for Protocol B. Although diagnostic labels remain excluded from model training, this threshold represents supervised calibration because validation labels determine the final operating point.

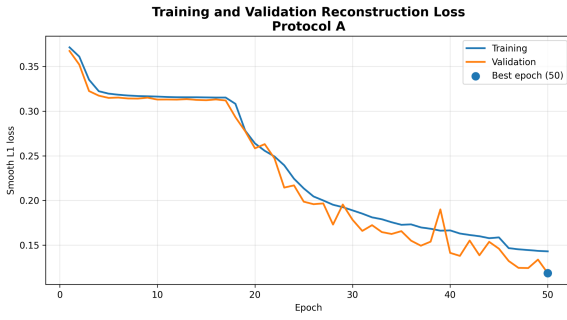
8 Experimental Results

8.1 Training Behaviour

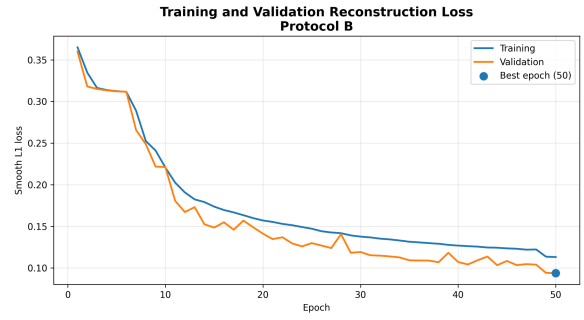
Both models completed 50 training epochs and achieved their lowest validation loss at the final epoch. Protocol B consistently produced lower reconstruction errors than Protocol A.

Table 6: Best validation reconstruction results.

Protocol	Smooth L1 loss	MAE	MSE
A	0.118472	0.245300	0.205431
B	0.093811	0.211621	0.147266

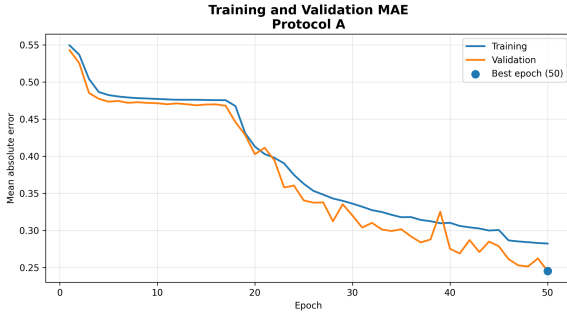


(a) Protocol A

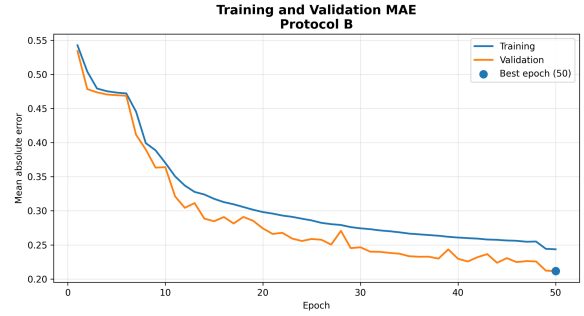


(b) Protocol B

Figure 2: Training and validation Smooth L1 reconstruction loss.



(a) Protocol A



(b) Protocol B

Figure 3: Training and validation mean absolute reconstruction error.

Validation errors are generally lower than training errors because dropout is active during training and disabled during evaluation. This behaviour should therefore not be interpreted as evidence of unusually strong generalization.

8.2 Recording-Score Distributions

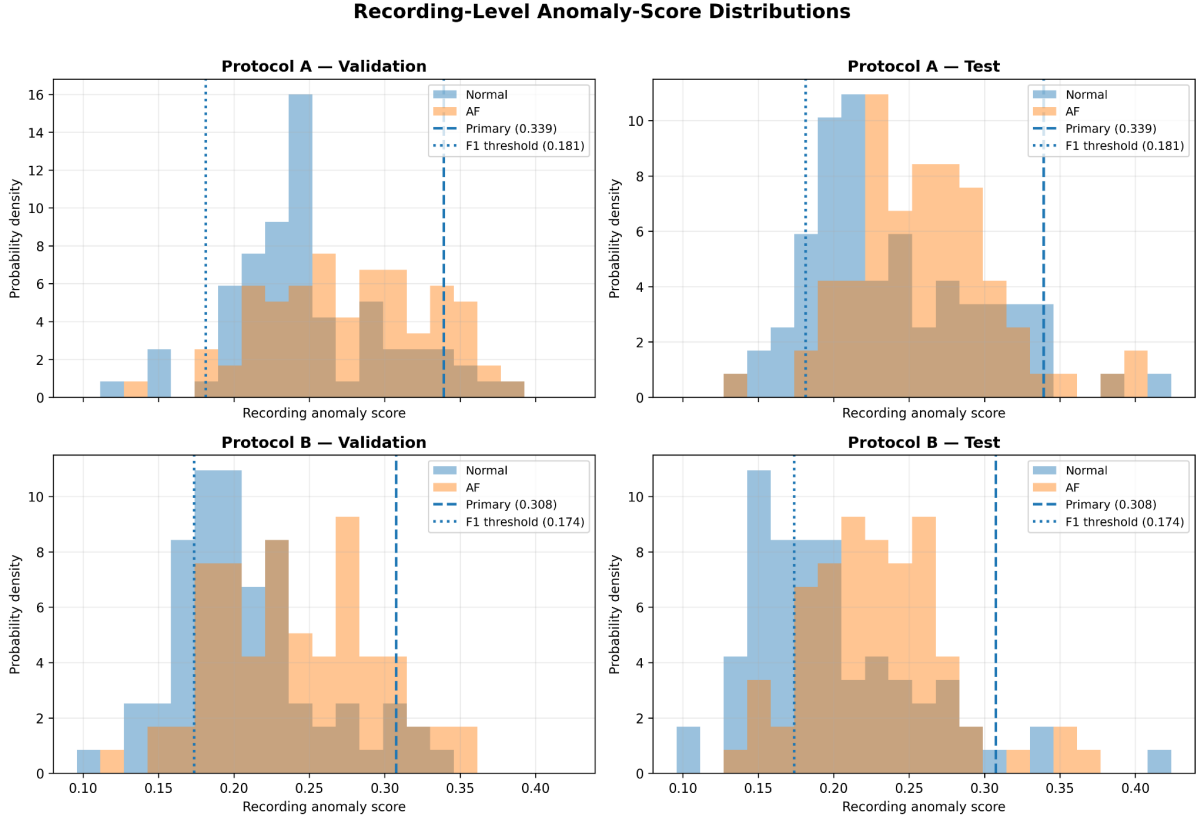


Figure 4: Recording-level anomaly-score distributions for validation and test data. AF recordings receive higher scores on average, especially under Protocol B, although substantial overlap remains between the two classes.

Table 7: Test recording-score distributions.

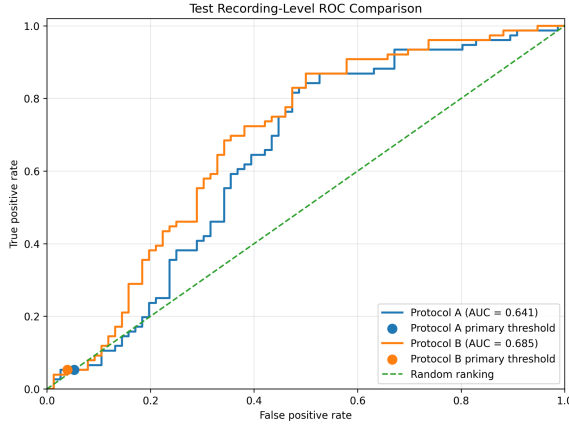
Protocol	Class	Count	Mean	Median	Std. dev.
A	Non-AF	76	0.239780	0.220274	0.058008
A	AF	76	0.261345	0.259348	0.047867
B	Non-AF	76	0.199747	0.184825	0.056696
B	AF	76	0.227800	0.224028	0.046126

For both protocols, AF recordings have higher mean and median anomaly scores. However, the overlapping distributions limit the ability of a single threshold to achieve both high sensitivity and high specificity.

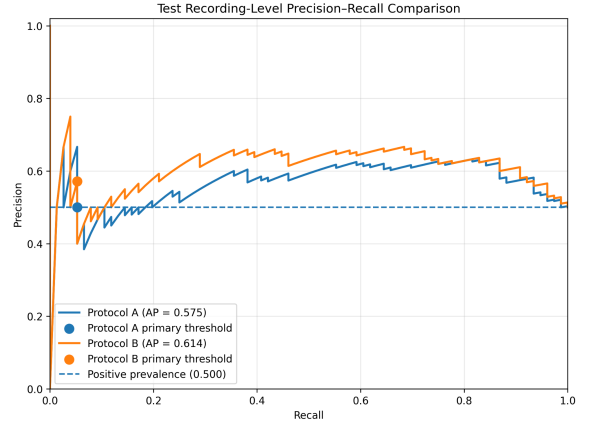
8.3 Threshold-Independent Evaluation

Table 8: Threshold-independent recording-level performance.

Protocol	Validation		Test	
	ROC-AUC	PR-AUC	ROC-AUC	PR-AUC
A	0.6569	0.6399	0.6413	0.5751
B	0.6827	0.6691	0.6851	0.6138



(a) ROC curves



(b) Precision-recall curves

Figure 5: Threshold-independent test comparison of the two protocols.

Protocol B achieves the strongest threshold-independent performance, with a test ROC-AUC of 0.6851 and PR-AUC of 0.6138. These results indicate improved ranking of AF recordings, although the overall discrimination remains moderate.

8.4 Threshold Sensitivity

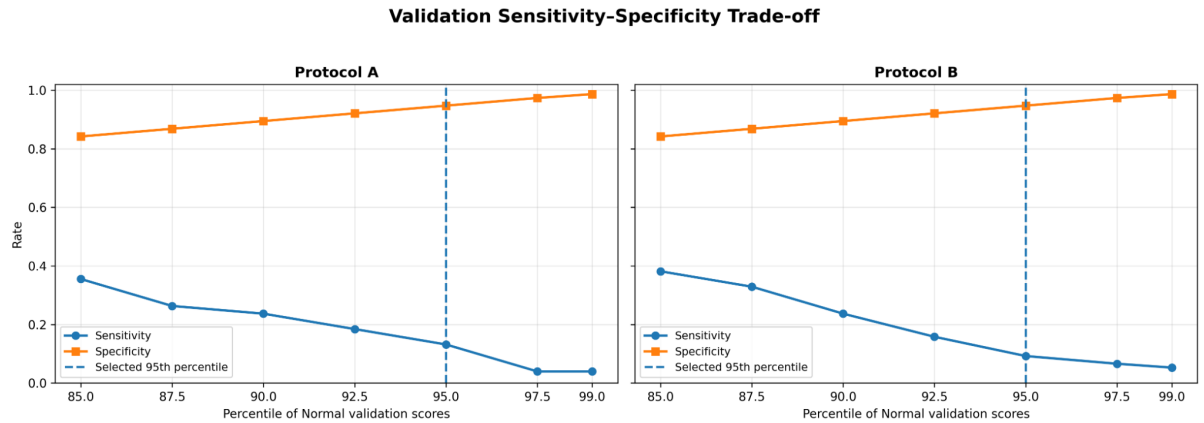


Figure 6: Validation sensitivity-specificity trade-off across percentiles of the non-AF validation-score distribution.

Increasing the percentile threshold reduces the number of false positives but also lowers AF sensitivity. The selected 95th-percentile threshold is therefore highly conservative.

8.5 Operating-Point Evaluation

Recording-Level Test Confusion Matrices

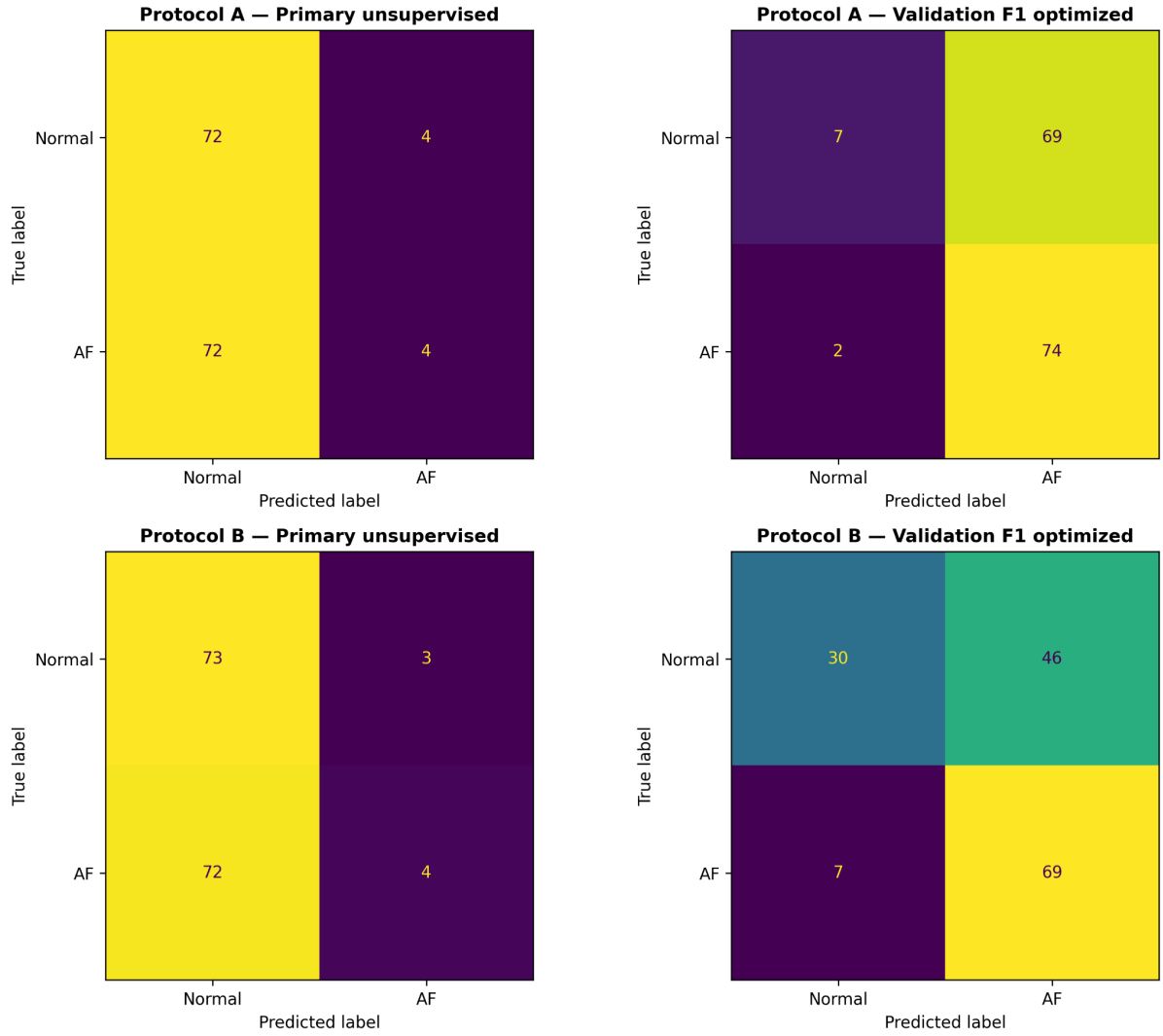


Figure 7: Recording-level test confusion matrices for both protocols and threshold strategies.

Table 9: Test performance at the primary percentile threshold.

Protocol	Bal. Acc.	Precision	Sensitivity	Specificity	F1	Threshold
A	0.5000	0.5000	0.0526	0.9474	0.0952	0.339208
B	0.5066	0.5714	0.0526	0.9605	0.0964	0.307567

Table 10: Test performance at the validation F1-optimized threshold.

Protocol	Bal. Acc.	Precision	Sensitivity	Specificity	F1	Threshold
A	0.5329	0.5175	0.9737	0.0921	0.6758	0.181348
B	0.6513	0.6000	0.9079	0.3947	0.7225	0.173709

The primary thresholds produce high specificity but detect very few AF recordings. In contrast, the F1-optimized thresholds substantially improve sensitivity and F1-score, at the cost of additional false positives. Protocol B provides the best compromise, reaching a balanced accuracy of 0.6513 and an F1-score of 0.7225.

8.6 Qualitative Reconstruction Analysis

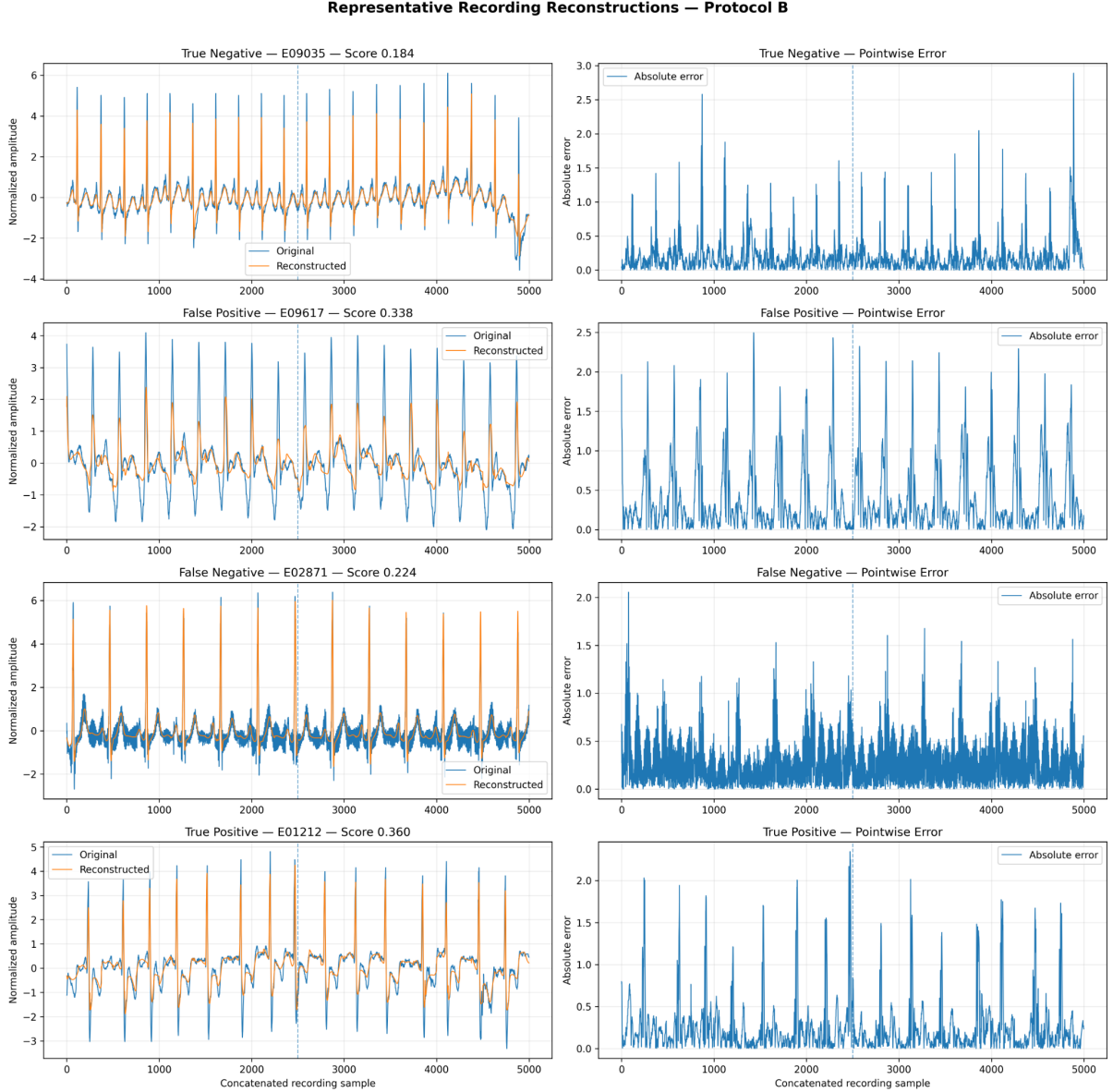


Figure 8: Representative Protocol B reconstructions and pointwise absolute errors for a true negative, false positive, false negative, and true positive under the primary threshold.

The examples confirm that reconstruction error is not specific to AF. Some non-AF recordings are reconstructed poorly and become false positives, while some AF recordings remain sufficiently reconstructable to fall below the conservative threshold.

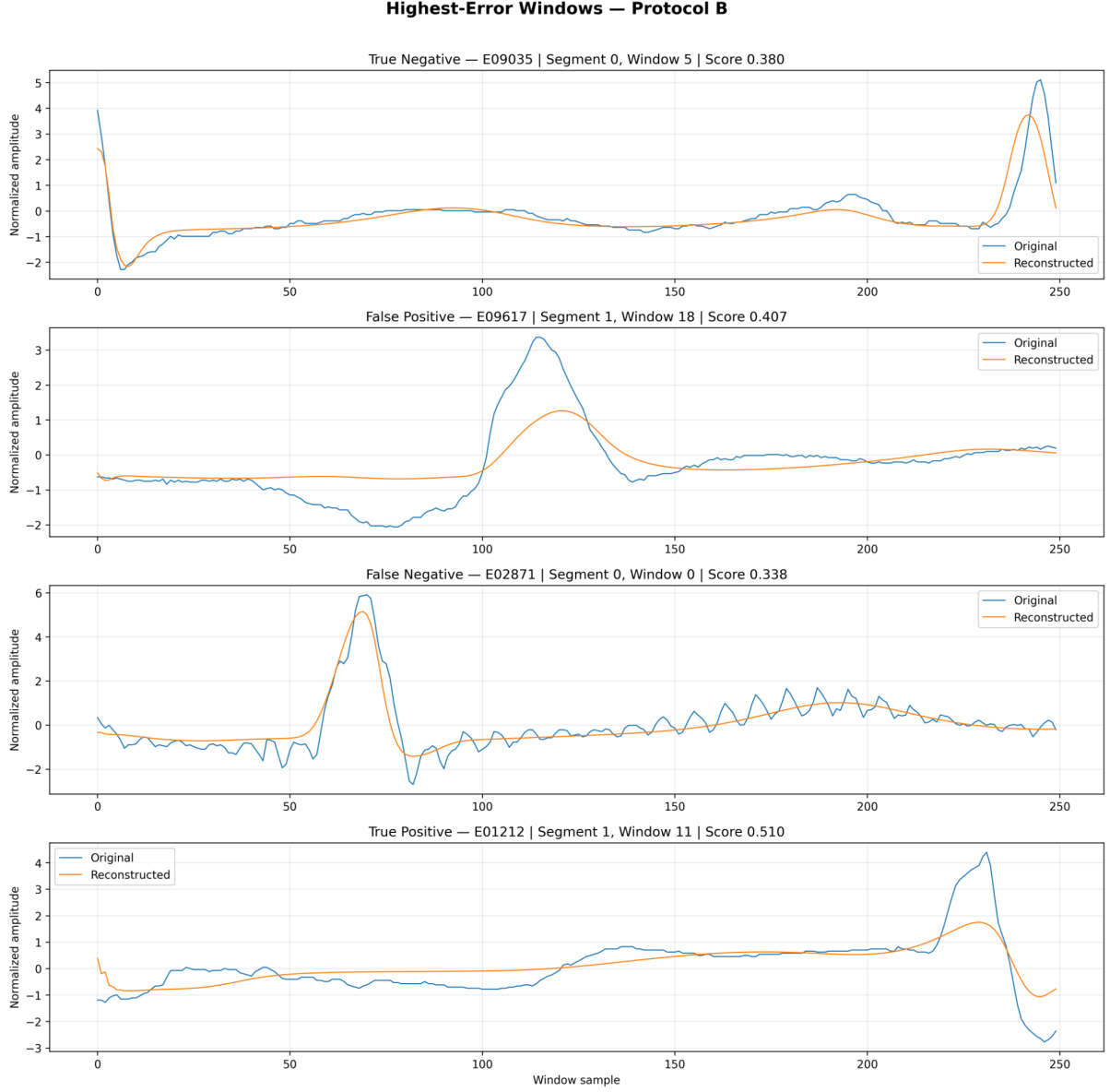


Figure 9: Highest-error windows for the representative Protocol B cases.

The highest-error windows provide a localized view of the waveform regions that contribute most strongly to each recording-level anomaly score.

8.7 Single-Recording Inference

Complete inference was demonstrated for AF recording E00005, which contains two segments and 38 windows. Protocol B produced segment-level scores of 0.247781 and 0.278659, resulting in a final recording score of

$$r_{E00005} = 0.278659.$$

Because

$$0.173709 < 0.278659 < 0.307567,$$

the recording is classified as normal by the primary threshold and anomalous by the F1-optimized threshold.

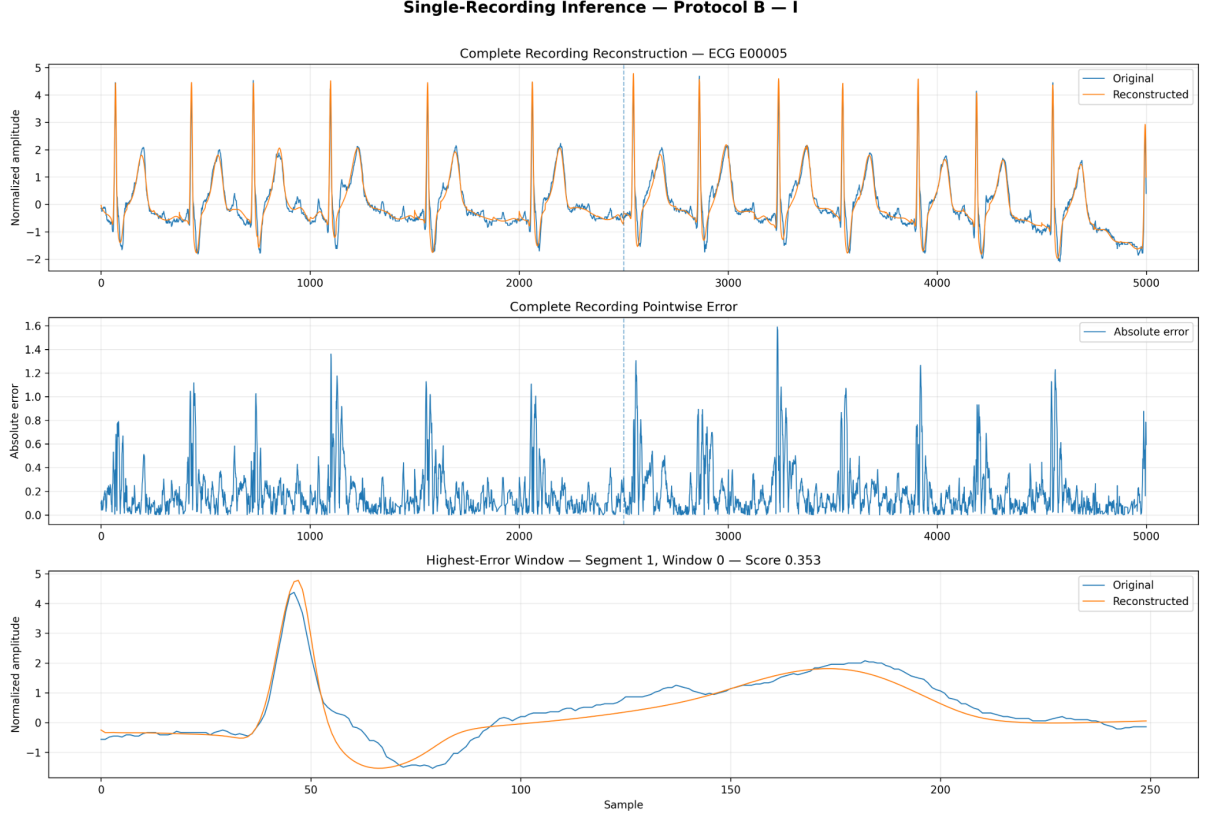


Figure 10: Unified Protocol B inference for AF recording E00005, including complete reconstruction, pointwise absolute error, and highest-error window.

9 Discussion

The experimental results show that Protocol B consistently outperforms Protocol A across the main evaluation metrics. Training the autoencoder exclusively on normal ECG recordings produces better separation between normal and AF recordings, resulting in higher ROC-AUC and PR-AUC values. This observation is consistent with the objective of reconstruction-based anomaly detection, where abnormal patterns are expected to yield larger reconstruction errors.

Although Protocol B achieved the best overall performance, the comparison should be interpreted with caution because it was trained using more recordings than Protocol A. Consequently, the observed improvement may be influenced by both the one-class training strategy and the larger amount of training data.

The threshold analysis highlights the trade-off between sensitivity and specificity. The primary unsupervised threshold provides high specificity but detects only a small proportion of AF recordings, whereas the validation-optimized threshold substantially improves sensitivity and the F1-score at the expense of more false positives. This demonstrates that threshold selection has a significant impact on the practical behaviour of the detector.

Overall, the implemented pipeline satisfies the objectives of the assignment by providing a complete, reproducible, and technically sound framework for reconstruction-based ECG anomaly detection. Future work could compare the proposed LSTM autoencoder with alternative architectures, such as GRU-, Transformer-, or convolutional autoencoders, while also investigating different window configurations, aggregation strategies, and larger ECG datasets.

10 Conclusion

This project implemented and evaluated a complete reconstruction-based ECG anomaly detection pipeline for the Georgia subset of the PhysioNet/Computing in Cardiology Challenge 2020 dataset using an LSTM autoencoder. The implemented workflow includes dataset validation, recording-level partitioning, training-only normalization, sliding-window generation, model training, hierarchical reconstruction-error aggregation, validation-based threshold selection, and recording-level evaluation.

Among the two experimental protocols, Protocol B consistently achieved the strongest overall performance, obtaining a test ROC-AUC of 0.6851 and a PR-AUC of 0.6138. These results indicate that training exclusively on non-AF recordings improves the separation between non-AF and AF recordings. Nevertheless, the considerable overlap between reconstruction-score distributions limits the effectiveness of threshold-based AF detection when simultaneously optimizing sensitivity and specificity.

Overall, the implementation satisfies the objectives of the assignment by providing a complete, reproducible, and technically sound baseline for reconstruction-based ECG anomaly detection. Future work may investigate alternative reconstruction architectures, different windowing and aggregation strategies, and evaluation on larger and more diverse ECG datasets.

A Learning-Rate Schedules

The `ReduceLROnPlateau` scheduler automatically reduced the learning rate from 10^{-3} to 5×10^{-4} during the final stage of training for both experimental protocols, as illustrated in Figure 11.

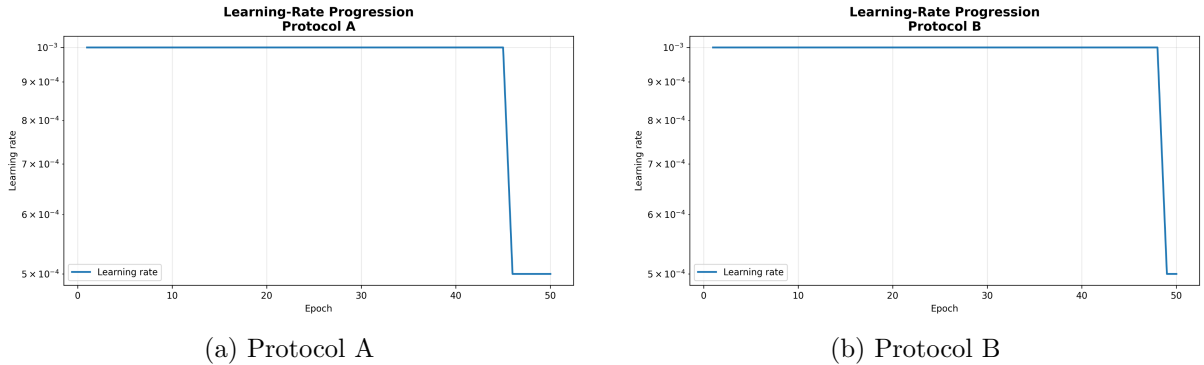


Figure 11: Learning-rate schedules for the two experimental protocols. The scheduler reduced the learning rate after the validation loss plateaued, allowing stable convergence during the final training epochs.

B Complete Final Comparison

Table 11 summarizes the principal training, validation, threshold-selection, and test results obtained for the two experimental protocols.

Table 11: Summary of the final notebook comparison.

Metric	Protocol A	Protocol B
Training recordings	1,068	2,340
Training windows	40,451	88,635
Best epoch	50	50

Metric	Protocol A	Protocol B
Best validation loss	0.1185	0.0938
Validation ROC-AUC	0.6569	0.6827
Validation PR-AUC	0.6399	0.6691
Primary threshold	0.3392	0.3076
F1-optimized threshold	0.1813	0.1737
Test ROC-AUC	0.6413	0.6851
Test PR-AUC	0.5751	0.6138
Primary balanced accuracy	0.5000	0.5066
Primary sensitivity	0.0526	0.0526
Primary specificity	0.9474	0.9605
Primary F1-score	0.0952	0.0964
F1-threshold balanced accuracy	0.5329	0.6513
F1-threshold sensitivity	0.9737	0.9079
F1-threshold specificity	0.0921	0.3947
F1-threshold F1-score	0.6758	0.7225

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